Machine Learning Lab Schedule

Experiments	Content	Week
1	**KNN** Algorithm for Iris Flower Classification	**Week 6** Firday： 6:45-10:05 PM Room T2510
2	Classification of Wine Dataset Using **PCA** and **SVM** Algorithms	**Week 8** Thursday： 10:30-12:15AM Thursday： 4:00-17:45 PM Room T2510
3	**CNN** for MNIST Digit Recognition	**Week 10** Wednesday： 4:00-17:45 PM Thursday： 4:00-17:45 PM Room T2510

Experiment 1

KNN Algorithm for Iris Flower Classification

2025

Objective

- To gain a fundamental understanding of the KNN algorithm
- To implement the KNN algorithm in Python using libraries such as scikit-learn
- To apply KNN in solving the iris flower classification problem
- To evaluate the performance of our trained KNN classification model

Environment

- Operating System: Windows 10
- Python programming environment:

  - Anaconda & Jupyter Notebook

Note: The lab computers have been installed with Anaconda.

- Import the required libraries:

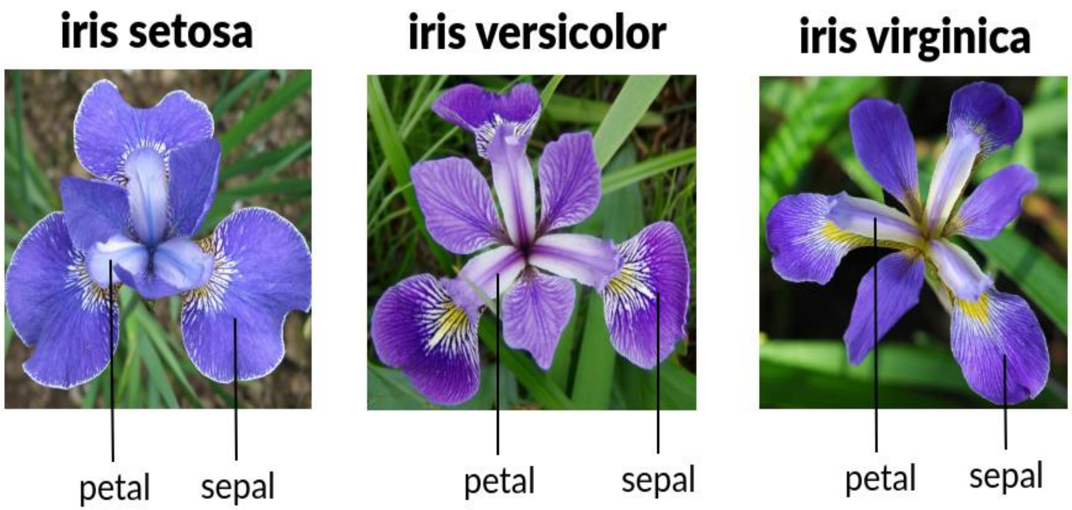
```python
import numpy as np  # NumPy library for numerical computations, including correlation calculation
import matplotlib.pyplot as plt # Matplotlib's plotting module for creating visualizations
```

Experiment Tasks

1. Dataset Loading

The Iris dataset consists of **50 samples** from **each of three species** of Iris. **Four features** were measured from each sample: the length and the width of the sepals and petals, in centimeters.

```
from sklearn import datasets #Load the Iris dataset from scikit-learn's datasets module
iris = datasets.load_iris()
```

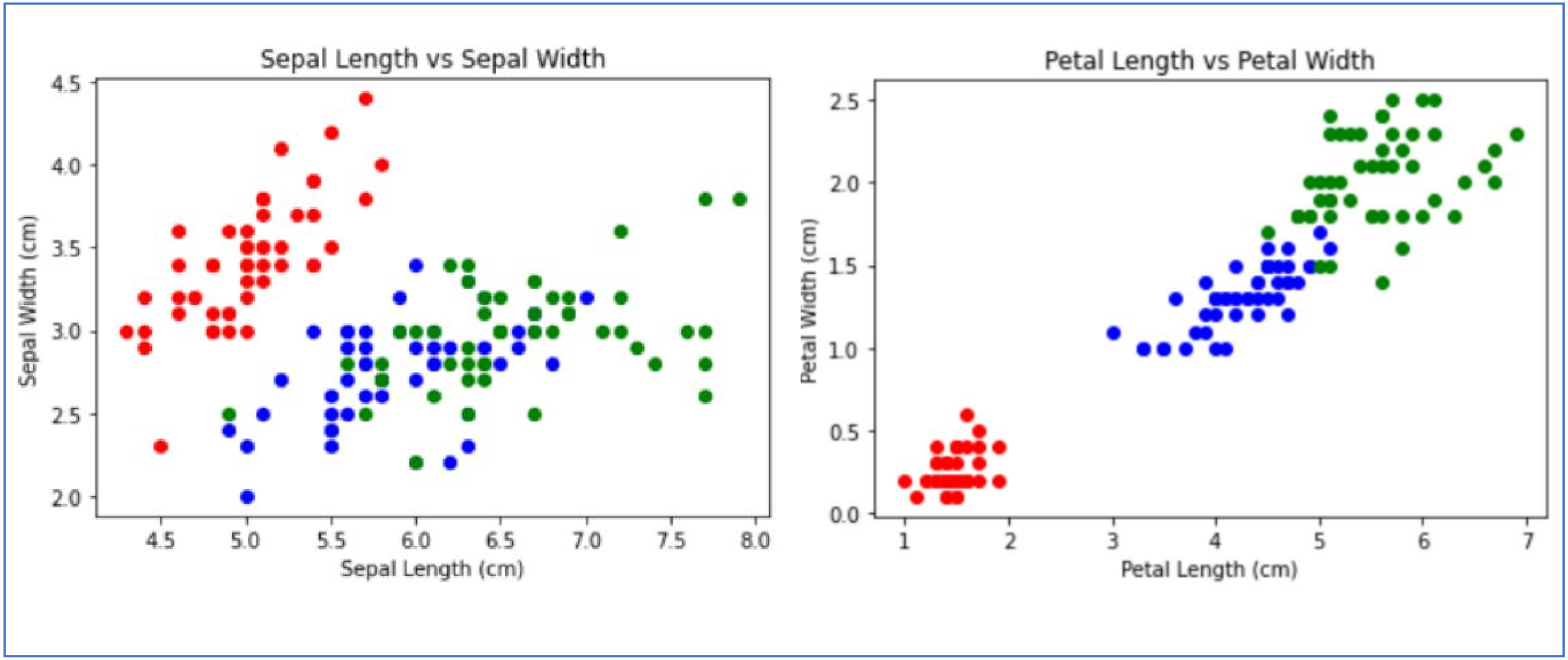


sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
5.1	3.5	1.4	0.2
4.9	3.0	1.4	0.2
4.7	3.2	1.3	0.2
4.6	3.1	1.5	0.2
5.0	3.6	1.4	0.2

Experiment Tasks

2. Data Exploration

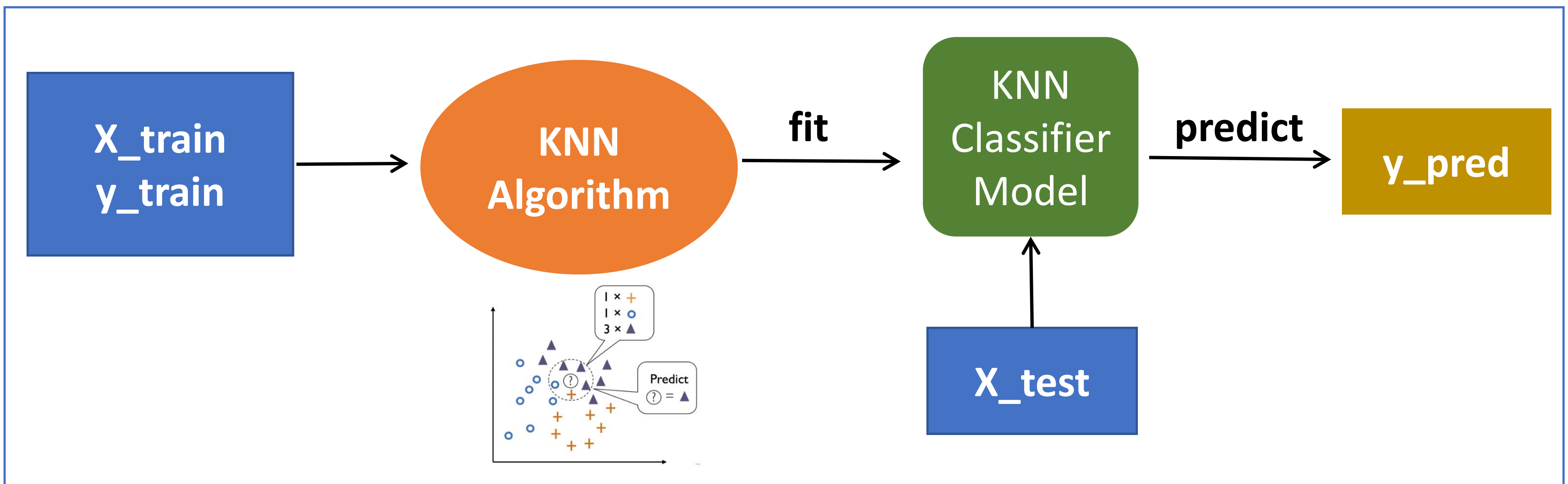
Use visualization tools to explore the relationship between sepal length and width, and petal length and width.



Experiment Tasks

3. Model Training

- Split the dataset into training and testing dataset
- Initialize and the KNN model with “n_neighbors=k”
- Pass the training dataset to KNN algorithm to train a model
- Pass the testing data to the trained model to predict the outcome



Experiment Tasks

4. Model Evaluation

- Evaluate the model's accuracy: compare the predictions against the actual outcomes

```
from sklearn.metrics import accuracy_score  # For evaluating model performance

# Print the overall accuracy score (fraction of correctly classified instances)
print("\nAccuracy Score:", accuracy_score(y_test, y_pred))
```

- Hyperparameter tuning: determine the best K.

THANK YOU!